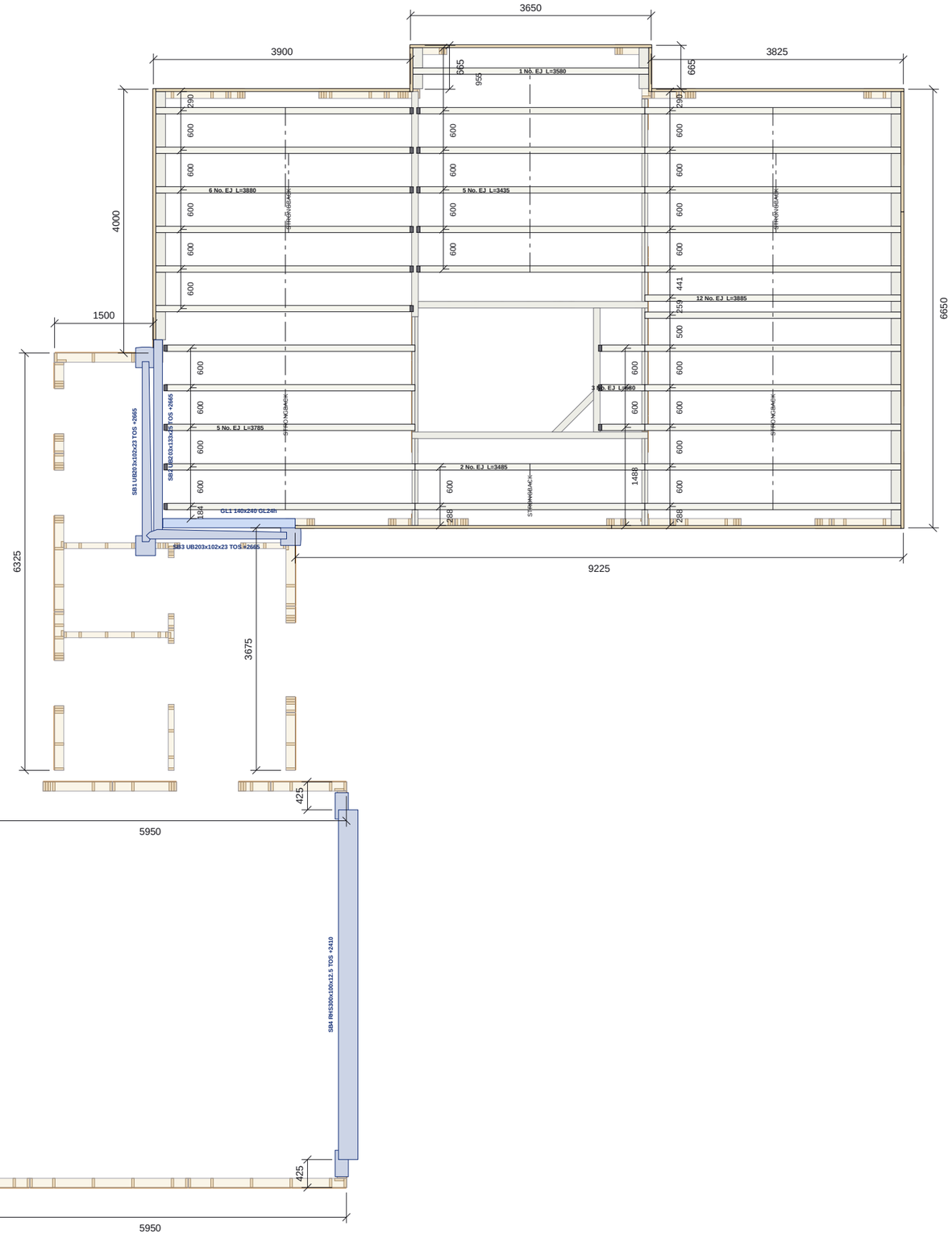


FIRST FLOOR STRUCTURE PLAN - INDICATIVE

Scale 1:75 @ A3 | Ground floor walls below as on the GA (studs cut at 1200); joists by Acorn Timber Engineering

0 1 2 3 4 5m

Do not scale. All dimensions in mm. Check all dimensions on site.

FIRST FLOOR STRUCTURE SCHEDULES

BEAMS

Mark	Section	Length	No.
GL1	140x240 GL24h	2008	1
SB1	UB203x102x23 S355JR (TLS)	2729	1
SB2	UB203x133x25 S355JR (TLS)	2955	1
SB3	UB203x102x23 S355JR (TLS)	2126	1
SB4	RHS300x100x12.5 S355J2H (TLS)	5300	1
PK	14mm marine ply packer, 100 wide	15.0m	6

PK: 14mm marine ply fixed to the top of each 240mm glulam to give the full 254mm floor depth (240 + 14 = 254, flush with the joists and edgebinder). Glulam and steel supplied by TLS; set out to faces as dimensioned; bearings to engineer. Steel TOS above underside of sole plate.

METAL-WEB JOISTS - BY ACORN

Type	Section	Length	No.
EJ	97x254 metal web	680	3
EJ	97x254 metal web	3435	5
EJ	97x254 metal web	3485	2
EJ	97x254 metal web	3580	1
EJ	97x254 metal web	3785	5
EJ	97x254 metal web	3880	6
EJ	97x254 metal web	3885	12

Acorn Timber Engineering supply, with 19 hangers, 5 strongback runs and noggins. Not in TLS procurement. Indicative; Acorn's layout supersedes.

EDGEBINDER, WAIST BAND, BREATHER

Item	Quantity	Order
Edgebinder 254 (45x195 + 45x59 rip)	32.3m	9 x 4.8m 45x195
45x59 C24 rips, 3 per 45x195	32.3m	3 x 4.8m 45x195
45x195 C24 total		12 x 4.8m
OSB/3 waist band 290x9	14 strips	4 x 1197x2397
Breather 450 x 2.7m strips, 150 lap	13 strips	14 (1 roll)

Run = modelled edgebinders. Timber cut-optimised +5%; OSB +7.5%; membrane laps +5%.

FLOOR DECKING

Item	Area	Order
22mm Egger P5 600x2400	73.6 m2	55 sheets
D4 adhesive 1L (per 6.5 sheets)		9 x 1L

Laid area from the model (27 full sheets + 49 cut pieces) = 51.1 sheets, +7.5%.

NOTES

1. INDICATIVE ONLY. Metal-web joists, hangers, strongbacks and noggins are designed and supplied by Acorn Timber Engineering; this drawing is superseded by Acorn's engineered floor layout on receipt.
2. Glulam beams 140x240 GL24h (dark blue) set out to the faces dimensioned, bearing on multi-ply studs to engineer's drawing 26-431-S-0010. 14mm marine ply fixed to the top of each 240mm glulam to give the full 254mm floor depth. Steel RHS300x100x12.5 / UB203x102x23 / UB203x133x25 beams and SHS100x100x6.3 / SHS120x120x10 posts supplied by TLS.
3. Edgebinder 254mm = 45x195 C24 with a 45x59 C24 rip fixed on top (rips cut 3 per 45x195, 3mm kerf).
4. OSB/3 waist band 290x9 to the outer face of the edgebinder, strips ripped 4 per 1197x2397 sheet.
5. Breather membrane over the waist band in 450 deep x 2.7m strips, each lapped 150 over the next.
6. 22mm Egger P5 flooring glued with D4 adhesive, fixed to Acorn's specification. Dimensions to frame and beam faces.

REVISIONS

REV	DATE	DESCRIPTION
C01	14.09.26	Construction issue
C02	14.09.26	Roof trusses revised; SP diagonals added
C03	11.10.26	FF windows coursed; steel + GP1 portal; steel drawings (ST)

TLS TIMBER FRAME
Design & Manufacture
Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE
FIRST FLOOR STRUCTURE - INDICATIVE

Beams, set-out and metal-web joist positions; Acorn layout supersedes

SCALE	SIZE	DATE	DRAWN
1:75	A3	11.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-FL-01	C03